

DareXAI – Full Stack AI Engineer Internship Challenge

Build an AI Business Operations Platform

Duration: 2 Days (48 Hours)

Objective

At DareXAI, we build AI-powered software that automates business operations using AI Agents, WhatsApp Automation, CRM Intelligence, Voice AI, and Workflow Automation.

Your challenge is to build a production-inspired **AI Business Operations Platform** that acts like an intelligent business assistant rather than a traditional dashboard.

The platform should enable business owners to:

- Converse naturally with an AI Agent
- Manage customers and opportunities
- Execute business actions using AI
- Automate follow-ups
- Monitor business activity
- Operate securely in a multi-tenant environment

We evaluate **engineering quality, architecture, security, AI integration, and problem-solving**—not the number of features you implement.

Core Requirements

Module	Requirements
Authentication & Security	Google OAuth (PKCE), Secure HTTP-only Cookies, JWT Access & Refresh Tokens, Refresh Token Rotation, Multi-Tenant Architecture, Tenant Isolation, Business Onboarding
AI Business Agent	Streaming Responses, Conversation History, Business Context Awareness, Tool Calling, AI Reasoning, Persistent Chat
CRM	Contacts, Opportunities, Opportunity Details, AI-generated Next Best Action
Unified Inbox	WhatsApp, Email and Call Logs displayed in a single conversation timeline with AI Summary, Sentiment and Intent Detection

WhatsApp Integration	Receive Webhooks, Store Messages, Display Conversations, Send AI-generated Replies (Meta Cloud API preferred, Sandbox acceptable)
Workflow Automation	Implement one complete AI workflow (e.g. Lead → AI Qualification → WhatsApp Follow-up → Task Creation → Audit Log)
Business Dashboard	Display live KPIs such as Active Opportunities, Revenue Pipeline, Pending Follow-ups, Customer Activity and AI Alerts
Security	Input Validation, Secure Cookies, CORS, SQL Injection Protection, XSS Protection, Environment Variables, Audit Logs
Testing	Authentication, Tenant Isolation, AI Tool Calling and one Frontend Component

AI Agent

The AI Assistant is the primary interface of the application.

It should be capable of:

Capability	Description
Conversational Interface	Respond naturally to user queries
Streaming	Stream responses token-by-token
Tool Calling	Invoke backend functions automatically
Context Awareness	Use onboarding information and CRM data
Explainability	Every AI response should include a short explanation of why it made that recommendation

Example:

"Follow up with Rahul tomorrow."

The AI should automatically:

1. Find Rahul in CRM
2. Generate a follow-up message
3. Send the message
4. Create a reminder
5. Store the action in the audit log

Note: Use Free Gemini API Key for this purpose, also use Streaming mode in text generation

Supported AI Tools

Implement at least the following tools:

Tool		Purpose
Search Contacts		Find customer information
Create Task		Create reminders or follow-ups
Update Opportunity		Update CRM status
Send WhatsApp		Send customer messages
Fetch Metrics	Business	Retrieve dashboard statistics

Unified Conversation Center

Create a unified communication hub.

Channel	Requirement
WhatsApp	Incoming & Outgoing Messages
Email	Email Timeline
Call Logs	Store and display call history

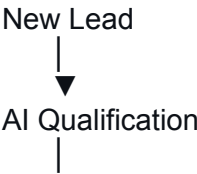
Every conversation should display:

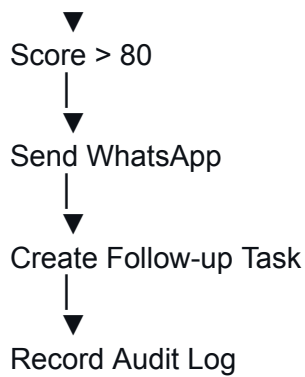
- AI Summary
- Intent Detection
- Sentiment
- Recommended Next Action

Workflow Automation

Implement one complete automation flow.

Example:





You may design your own workflow provided it demonstrates real business automation.

WhatsApp Integration (Bonus)

Implement a production-ready integration layer.

Feature	Requirement
Incoming Webhooks	Receive customer messages
Message Storage	Persist all conversations
Conversation UI	Display complete chat history
AI Reply	Generate and send responses
API	Meta Cloud API preferred (Sandbox acceptable)

If API credentials are unavailable, your implementation should require only environment variable changes to become operational.

You can use open source framework like <https://github.com/ArnasDon/wacrm> Whatsapp CRM

Data Model

Your application should include (minimum):

Entity

- Tenants
- Users
- Contacts
- Opportunities
- AI Chatbot Messages
- AI Chatbot Conversations
- Refresh Tokens
- Audit Logs

You may extend the schema as needed. Explain your design decisions in the README.

Recommended Technology Stack

Layer	Recommendation
Frontend	Next.js 14+
Backend	Next.js API / Node.js / Django
Database	PostgreSQL and Mongo
ORM	Prisma
AI	Gemini / OpenAI / Claude (prefer Use free gemini tier)
Authentication	Google OAuth
State Management	React Query + Zustand

Alternative technologies are acceptable if justified.

The demo should cover:

- Login
- Business Onboarding
- AI Conversation

- Basic CRM WhatsApp / Email Action
- Workflow Automation
- Multi-Tenant Isolation
- AI Reasoning
- Architecture Walkthrough

We have also attached the demo UI in this tab can refer - https://docs.google.com/document/d/10oXQColyVfB9PclRC29UV_Dwp_UPI3tGO-URvFyUunU/edit?tab=t.cwtdu4679bn8 (Note its just for the reference, you are advised to bring your own sense in it)

Submission Instructions

GitHub Repository

1. Create a Private GitHub Repository.

Follow and Grant access to:

GitHub Username:

addygeek and darexai (darexai06@gmail.com)

Include:

- Complete source code with Readme and setup
- Architecture explanation

2. Record Demo Video [MUST]

Upload the demonstration video to: YouTube (Unlisted) OR Google Drive or Loom link

Share the accessible link in the submission form.

Submission form link-

<https://docs.google.com/forms/d/e/1FAIpQLScBNo0OqfJqbcblhMKfmjlhvvyWNCAAf0kOWcwfs2fHZBuo b7g/viewform?usp=sharing&oid=109456902429399034535>

Follow us on - <https://www.linkedin.com/company/dare-xai/> , <https://www.linkedin.com/company/dare-xai/>

AI Usage Policy

You are encouraged to use AI tools such as ChatGPT, Gemini, Claude, Cursor, GitHub Copilot, or similar during development.

However, you must be able to explain every architectural and engineering decision in your submission.